PH101: 2023

Special Theory of Relativity
Lecture-2

Subhaditya Bhattacharya

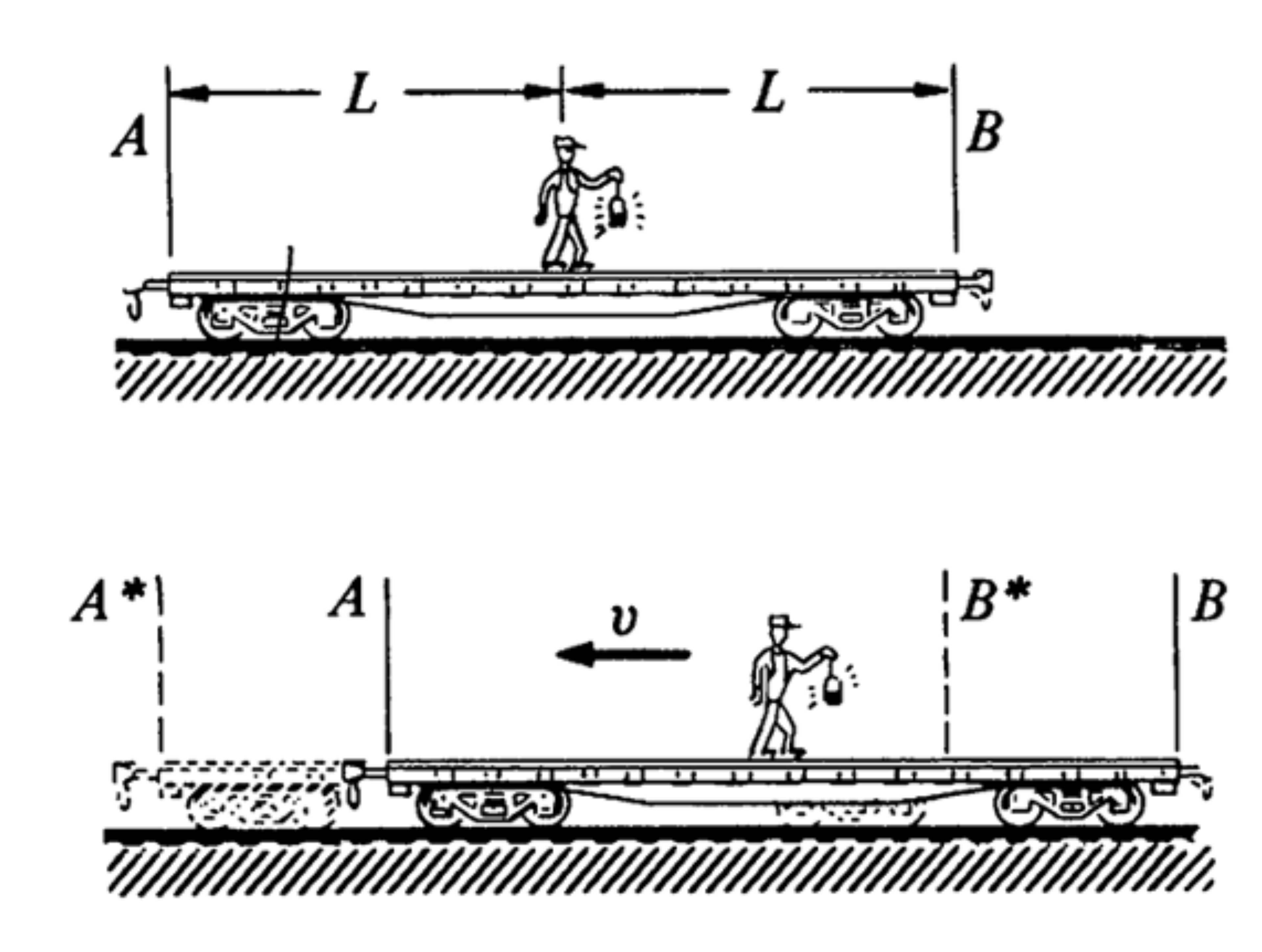
Recap

- Galilean Transformation (GT) to relate space-time of events in inertial frames
- Invariance of Newton's laws under GT
- Non-invariance of Maxwell's equations under GT
- Michelson Morley Experiment and null result
- Absence of ether and constancy of velocity of light
- Einstein's postulates of Special Theory of Relativity (STR)

We need to find coordinate transformations that comply with STR

Concept of Simultaneity

Two events are simultaneous then the time coordinates of both the events will be the same.



However, simultaneity in one coordinate system doesn't necessarily indicate the events to be simultaneous in another one.

Consider the following example:

In the rest frame, light pulse reaching A and B *simultaneously.*

$$t_L = t_R = \frac{L}{c}$$

However, in the moving frame, the light pulse reaches B* earlier than A*.

$$t_R = \frac{L}{c+v}, \; t_L = \frac{L}{c-v}$$

Therefore the concept of simultaneity is frame dependent even in Galilean transformation.

Homogeneity of space and time

Assumption: **space** and **time** are **homogeneous** (all points in space and time are equivalent) means, for example, that the results of a measurement of a length or time interval of a specific event should not depend on where or when the interval happens to be in our reference frame.

Let $x' = a_{11} x^2$. Then $x_2' - x_1' = a_{11}(x_2^2 - x_1^2)$.

In S-frame : If $x_1 = 1$, $x_2 = 2$, then in S' frame: $x_2' - x_1' = 3a_{11}$.
If $x_1 = 4$, $x_2 = 5$, then in S' frame: $x_2' - x_1' = 9a_{11}$.

That is, the measured length of the rod would depend on where it is in space. Likewise, we can reject any dependence on t that is not linear, for the time interval of an event should not depend on the numerical setting of the hands of the observer's clock.

$$
\begin{aligned}
x' &= a_{11}x + a_{12}y + a_{13}z + a_{14}t \\
y' &= a_{21}x + a_{22}y + a_{23}z + a_{24}t \\
z' &= a_{31}x + a_{32}y + a_{33}z + a_{34}t \\
t' &= a_{41}x + a_{42}y + a_{43}z + a_{44}t.
\end{aligned}
$$

The relationship between space time coordinates must be linear, not to provide a physical preference of one point over the others.

Lorentz Transformation

Following homogeneity and isotropy:

$$\begin{aligned} x' &= a_{11}x + a_{12}y + a_{13}z + a_{14}t \\ y' &= a_{21}x + a_{22}y + a_{23}z + a_{24}t \\ z' &= a_{31}x + a_{32}y + a_{33}z + a_{34}t \\ t' &= a_{41}x + a_{42}y + a_{43}z + a_{44}t \end{aligned}$$

x axis coincides with x' axis. This is so, when,

$$(y = 0, z = 0) \rightarrow (y' = 0, z' = 0)$$

$$\boxed{\begin{aligned} y' &= a_{22}y + a_{23}z \\ z' &= a_{32}y + a_{33}z \end{aligned}}$$

$$a_{21} = a_{24} = a_{31} = a_{34} = 0$$

- (x, y) plane $\rightarrow (x', y')$ plane.

$$z = 0 \rightarrow z' = 0$$

- (x, z) plane $\rightarrow (x', z')$ plane.

$$y = 0 \rightarrow y' = 0$$

$$\boxed{\begin{aligned} y' &= a_{22}\ y \\ z' &= a_{33}\ z \end{aligned}} \qquad a_{23} = a_{32} = 0$$

- Using principle of relativity,

$$\boxed{\begin{aligned} y' &= y \\ z' &= z \\ x' &= a_{11}(x - vt) \\ t' &= a_{41}x + a_{44}t \end{aligned}} \qquad x' = 0 \rightarrow x = vt$$

Symmetry argument

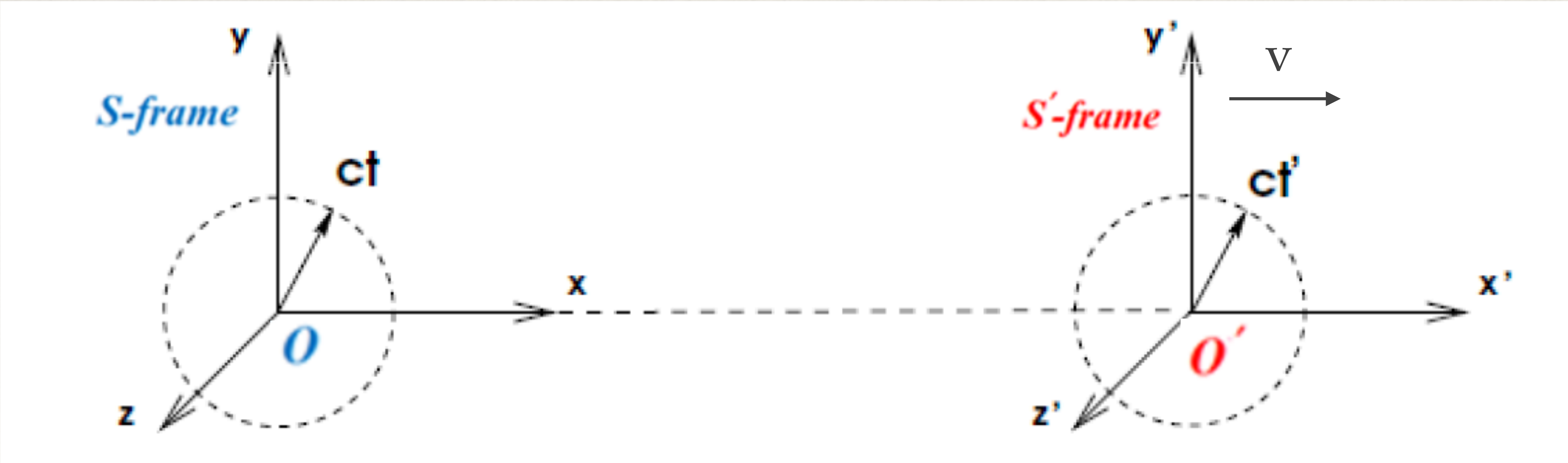
$$x' = a(x - vt) \quad and \quad t' = b(t + fx)$$

Lorentz transformations

S' moving with velocity v relative to S.

The origins coincide at $t = t' = 0$ and at this instant a burst of light is released.

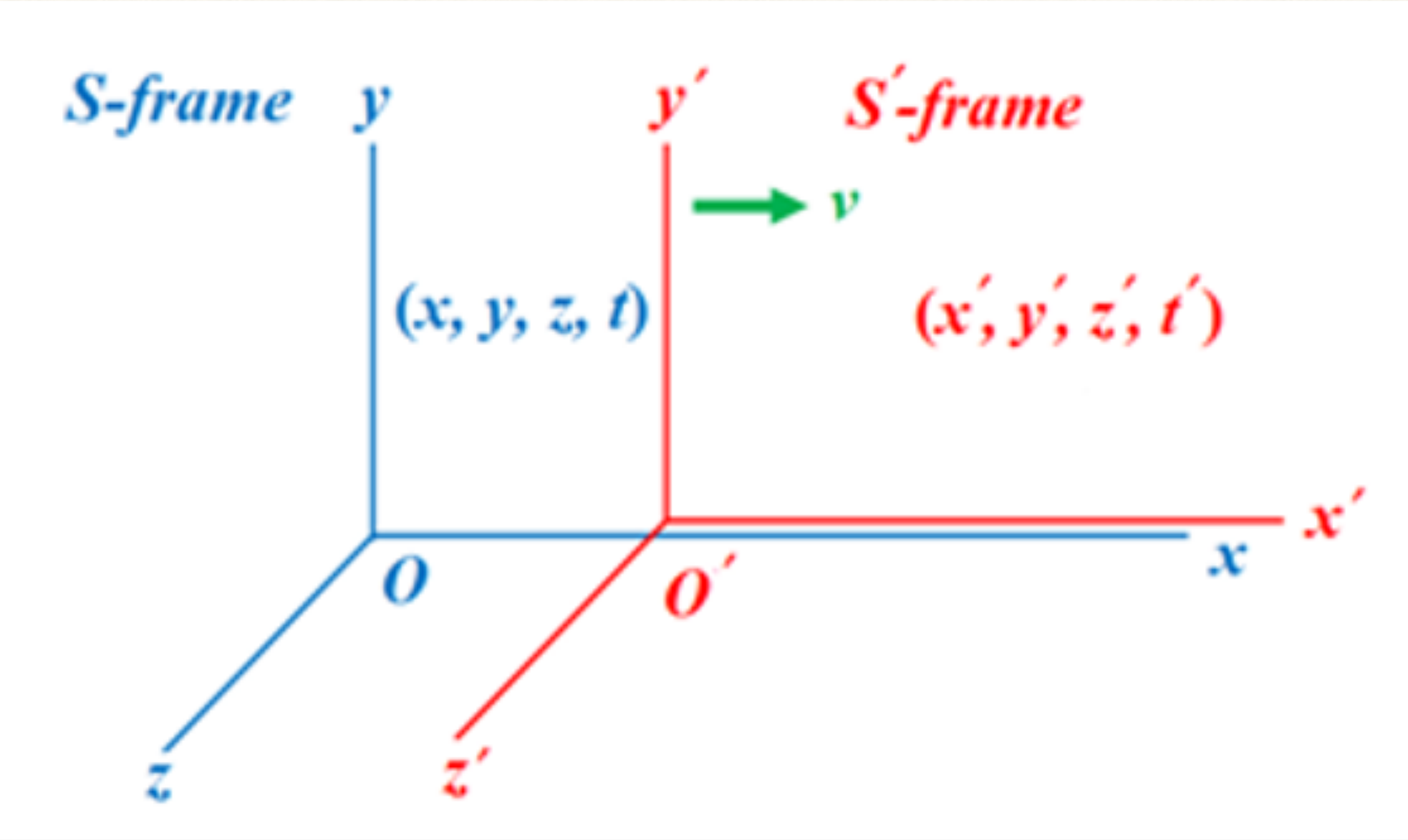
Speed of light is c in both frames.



In S, the wavefront of the light forms a sphere of radius $r = ct$ centred at O.

In S', the wavefront of the light forms a sphere of radius $r = ct'$ centred at O'.

$$x^2 + y^2 + z^2 = c^2t^2 \qquad and \qquad x'^2 + y'^2 + z'^2 = c^2t'^2$$



Question:

How (x, y, z, t) and (x', y', z', t') are related?

Lorentz Transformation: Continued

Following, $x'^2 + y'^2 + z'^2 = c^2 t'^2$

$$\therefore \qquad a^2(x - vt)^2 + y^2 + z^2 = c^2 b^2 (t + fx)^2$$

Rearranging:

$$x^2(a^2 - f^2b^2c^2) - 2xt(a^2v + f^2b^2c^2) + y^2 + z^2 = c^2t^2(b^2 - \frac{a^2v^2}{c^2})$$

However, this is equivalent to: $x^2 + y^2 + z^2 = c^2t^2$

$$a^2 - f^2b^2c^2 = 1; \qquad a^2v + f^2b^2c^2 = 0; \qquad b^2 - \frac{a^2v^2}{c^2} = 1$$

This can be solved to show:

$$a = b = \frac{1}{\sqrt{1-\beta^2}} = \gamma; \qquad \beta^2 = \frac{v^2}{c^2}, \quad and \quad f = -\frac{v}{c^2}$$

$$x' = \gamma(x - vt), \quad y = y', \quad z = z', \quad and \quad t' = \gamma(t - \frac{v}{c^2}x)$$

Lorentz's Transformations	Inverse Lorentz's Transformations
$x' = \dfrac{x - vt}{\sqrt{1 - v^2/c^2}}$	$x = \dfrac{x' + vt'}{\sqrt{1 - v^2/c^2}}$
$y' = y$	$y = y'$
$z' = z$	$z = z'$
$t' = \dfrac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}$	$t = \dfrac{t' + (v/c^2)x'}{\sqrt{1 - v^2/c^2}}$

Lorentz transformations are valid for any *v*.

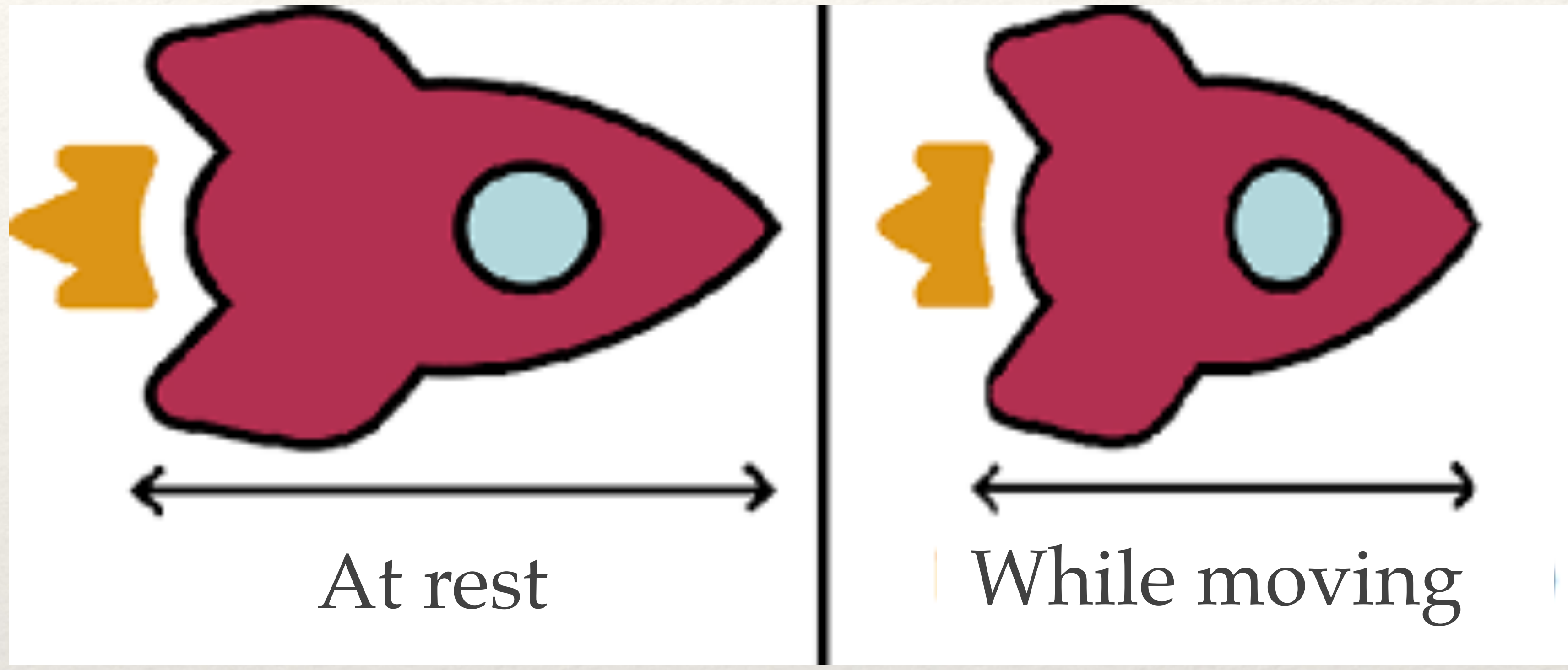
For v << *c*, they reduce to the Galilean transformations.

v > *c* produces imaginary, non-physical numbers, *c* becomes a limiting speed for physical phenomena.

Lorentz Transformation:consequences

Length Contraction:

A body's length is measured to be the greatest when it is at rest relative to the observer. When it moves with a velocity v relative to the observer, its measured length is contracted in the direction of its motion, whereas the its dimension perpendicular to the direction of motion remains unaffected.

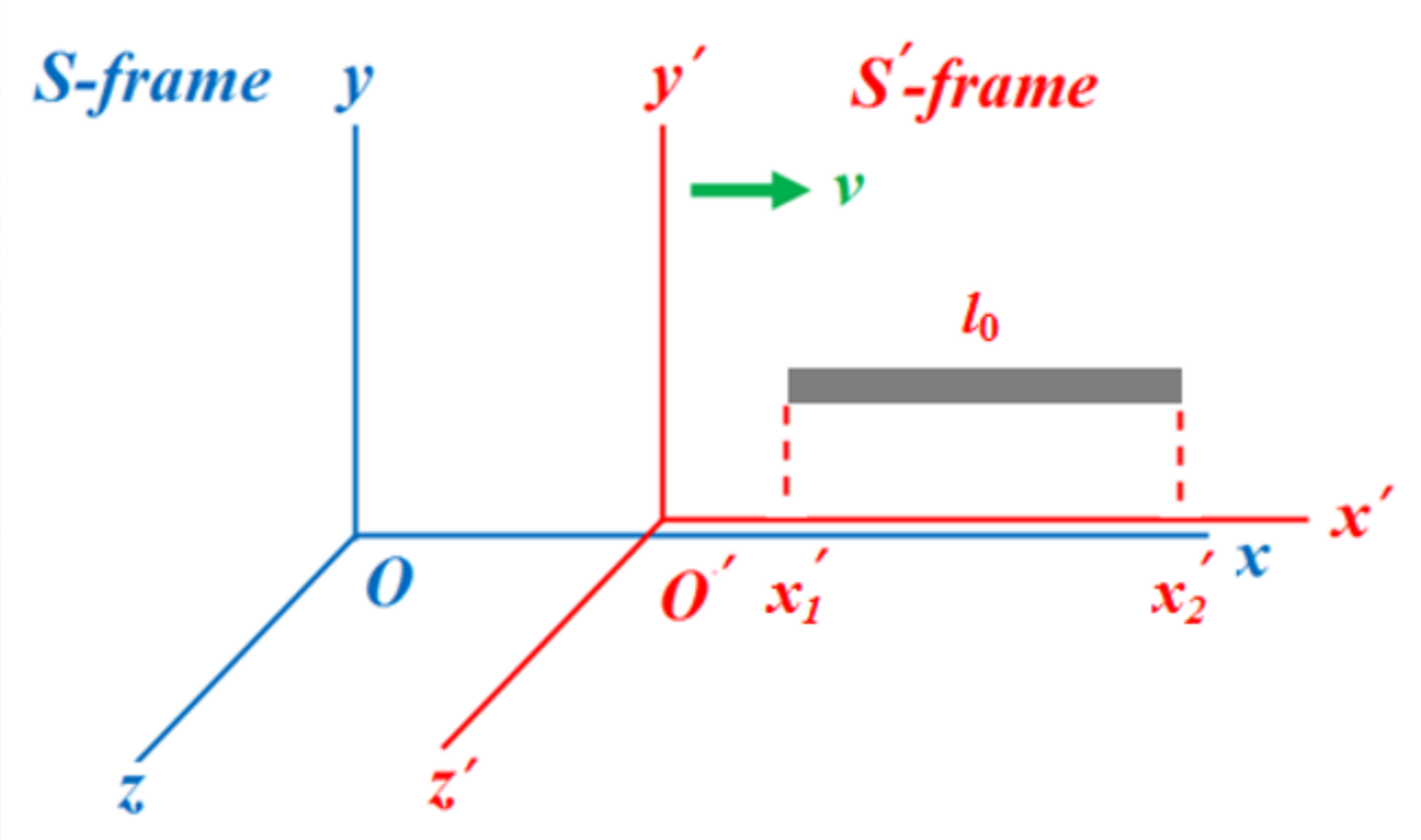


Time dilation

A clock is measured to go at its fastest rate when it is at rest relative to the observer. When it moves with a velocity v relative to the observer, its rate is measured to have slowed down.

Length Contraction

Imagine a rod lying at rest along the x'-axis of the S'-frame. Its end points are measured to be at x_2' and x_1', so that its rest length is $x_2' - x_1'$. What is the rod's length as measured by the S-frame observer, for whom the rod moves with the relative speed v? From the first Lorentz equation, we have



$$x_2' = \frac{x_2 - vt_2}{\sqrt{1-\beta^2}} \qquad x_1' = \frac{x_1 - vt_1}{\sqrt{1-\beta^2}}$$

$$x_2' - x_1' = \frac{(x_2 - x_1) - v(t_2 - t_1)}{\sqrt{1-\beta^2}}.$$

The length of the rod measured in the S frame is between the two end points measured at same time.

$$t_2 = t_1$$

The frame in which the observed body is at rest is called the **proper frame** and the length is called **proper length** (l_0)

$$x_2' - x_1' = \frac{x_2 - x_1}{\sqrt{1-\beta^2}}$$

$$x_2 - x_1 = (x_2' - x_1')\sqrt{1-\beta^2}$$

$$x_2' - x_1' = l_0$$

$$x_2 - x_1 = l$$

$$l = \frac{l_0}{\gamma}$$

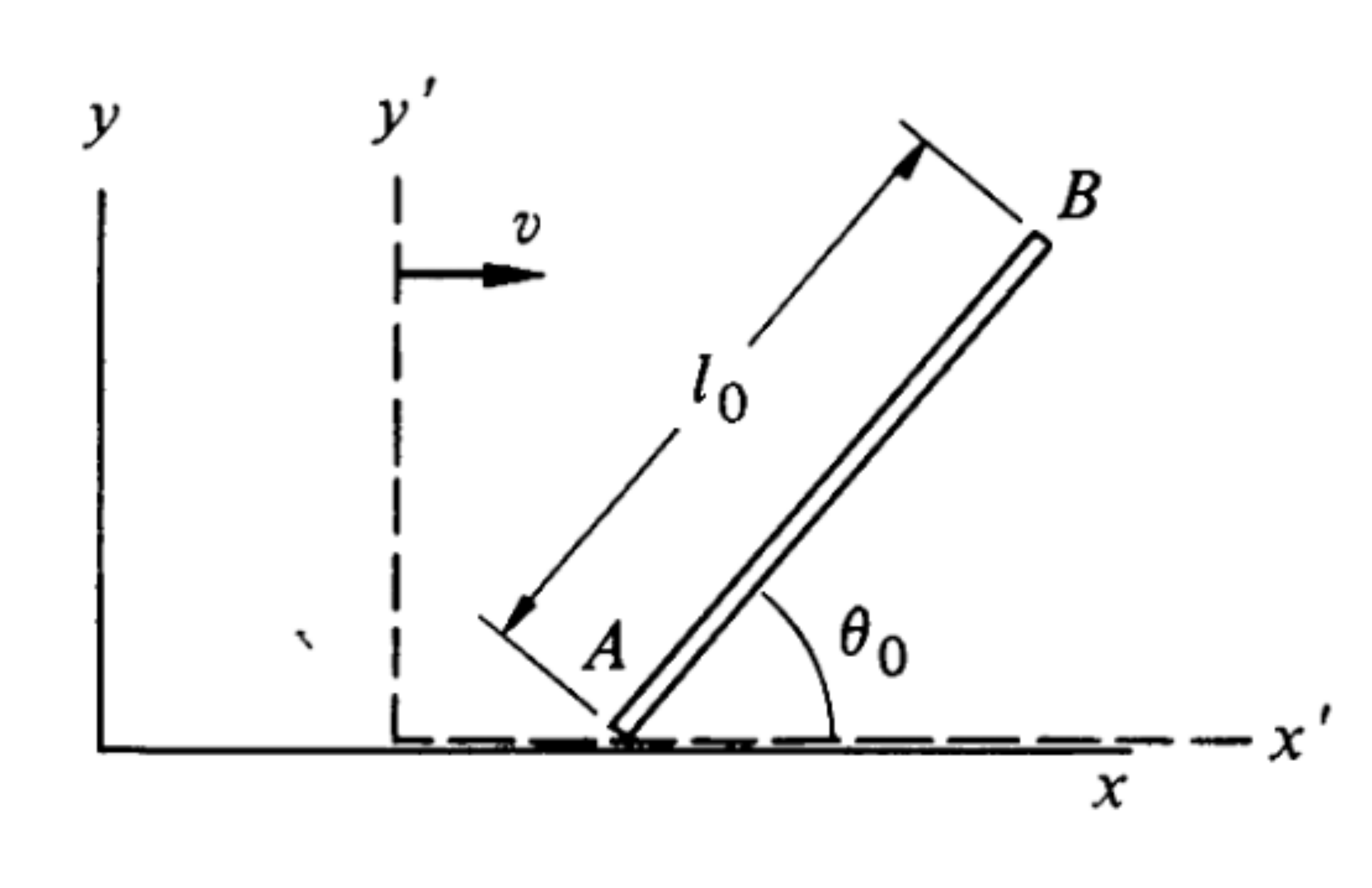
$$l < l_0$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} > 1$$

Recall, the Length contraction could explain the null result in Michelson Morley experiment

Example

A rod of length l_0 lies in the $x'y'$ plane of its rest system and makes an angle θ_0 with the x' axis. What is the length and orientation of the rod in the lab system x, y in which the rod moves to the right with velocity v?



In (x', y') frame:

A:	$x'_A = 0$	$y'_A = 0$
B:	$x'_B = l_0 \cos\theta_0$	$y'_B = l_0 \sin\theta_0.$

Using Lorentz transformation:

$$A:\ x'_A = 0 = \gamma(x_A - vt)$$
$$B:\ x'_B = l_0 \cos\theta_0 = \gamma(x_B - vt)$$
$$A:\ y'_A = 0 = y_A$$
$$B:\ y'_B = l_0 \sin\theta_0 = y_B.$$

The angle that the rod makes with the x axis is

$$\theta = \arctan\frac{y_B - y_A}{x_B - x_A} = \arctan\left(\gamma\frac{\sin\theta_0}{\cos\theta_0}\right) = \arctan(\gamma\tan\theta_0).$$

The moving rod is both contracted and rotated.

$$x_B - x_A = \frac{l_0\cos\theta_0}{\gamma}$$
$$y_B - y_A = l_0 \sin\theta_0.$$

The length is

$$l = [(x_B - x_A)^2 + (y_B - y_A)^2]^{\frac{1}{2}} = l_0\left[\left(1 - \frac{v^2}{c^2}\right)\cos^2\theta_0 + \sin^2\theta_0\right]^{\frac{1}{2}} = l_0\left[1 - \frac{v^2}{c^2}\cos^2\theta_0\right]^{\frac{1}{2}}.$$

Time Dilation

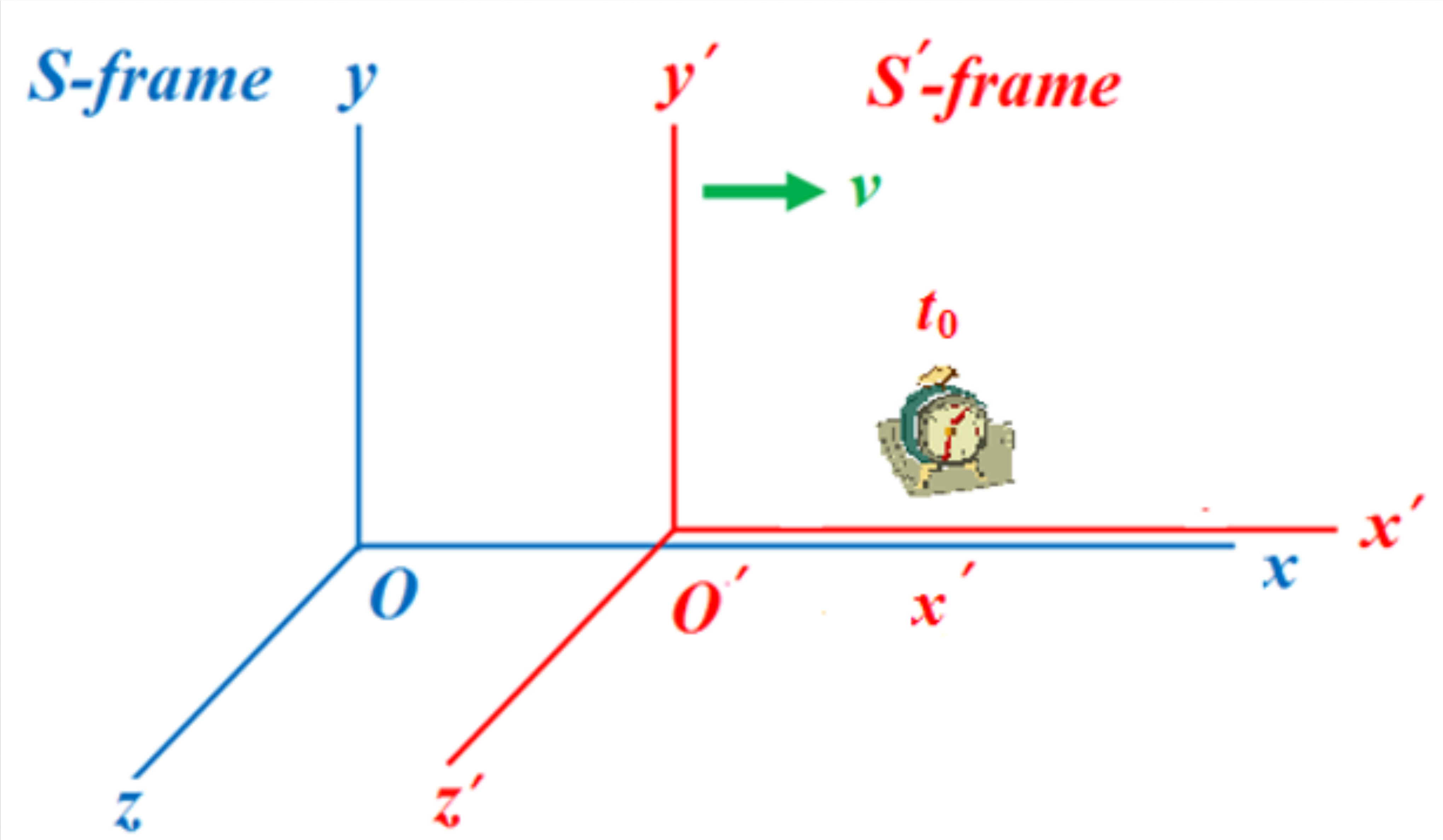
Consider a clock at rest in (x', y') system.

Consider two events A and B measured at same x'_0

$A: \; x'_0 \quad t'_A$

$B: \; x'_0 \quad t'_B$

$\tau = t'_B - t'_A$ **Proper time**



Question: What will be the interval in S frame ?

Using inverse Lorentz transformation: $t = \gamma(t' + \frac{vx'}{c^2})$

$$t_A = \gamma\left(t'_A + \frac{vx'_0}{c^2}\right)$$

$$t_B = \gamma\left(t'_B + \frac{vx'_0}{c^2}\right).$$

$$T = \gamma(t'_B - t'_A)$$
$$= \gamma\tau$$
$$= \frac{\tau}{\sqrt{1 - v^2/c^2}}.$$

The time interval in the laboratory system is greater than the rest system. The moving clock runs slow!!

Defining

$T = t_B - t_A$

$T > \tau$

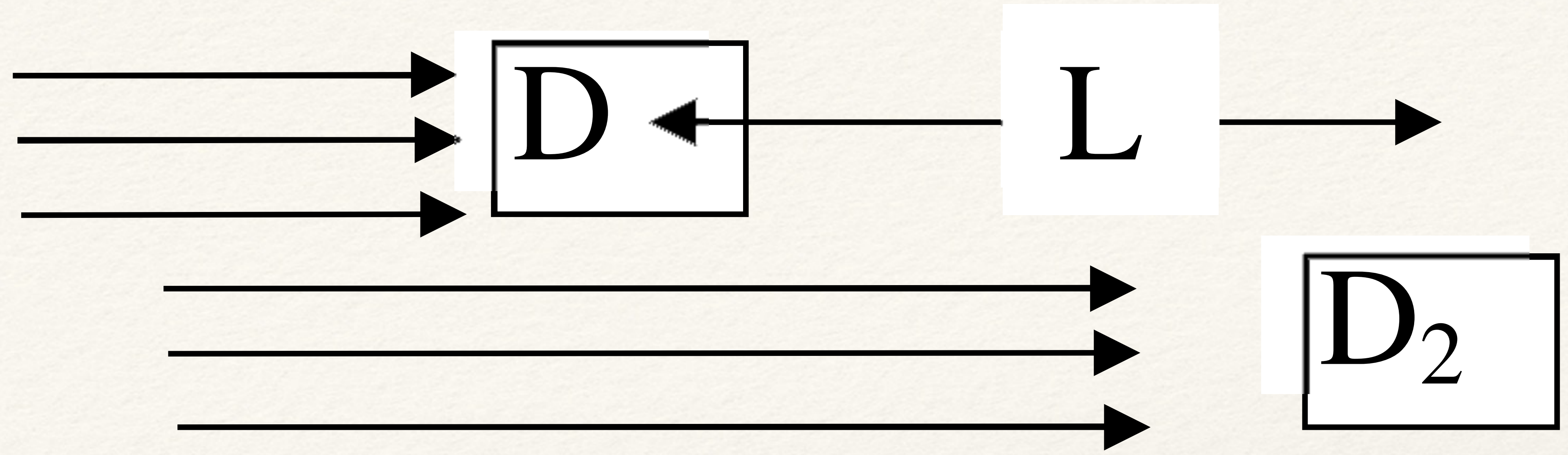
$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} > 1$$

Time dilation visualisation

Twin paradox

Example

Example: A beam of identical unstable particles flying at a speed v is sent through two detectors separated by a distance L. It is observed that N_1 particles are recorded at the first counter (D) and N_2 at the second detector (D_2), the reduction being solely due to the decay of the particles in flight.



Find the lifetime (τ_0) of the particle (in its rest frame).

Let ΔT be the time taken by the particles to reach D_2 from D, w.r.t the ground observer: $\Delta T = \dfrac{L}{v}$

Now, $$N_2 = N_1 e^{-\Delta T/(\gamma\tau_0)} = N_1 e^{-L/(v\gamma\tau_0)}$$

$$\ln\left(\frac{N_1}{N_2}\right) = \frac{L}{v\gamma\tau_0}$$

$$\boxed{\tau_0 = \frac{L}{\ln\left(\frac{N_1}{N_2}\right) c\sqrt{\gamma^2 - 1}}}$$

$\gamma\tau_0$: lifetime of the particle w.r.t the observer

Consider the particles mentioned are muons. We can then determine the lifetime of muons (at rest), in an experiment knowing

(i) their speed (say, $c\sqrt{8/3}$) through the detector,

(ii) $L = 200$ m

(iii) N_1 and N_2 were measured to be 10,000 and 8,983, in D1 and D2 respectively.

$$\gamma = \frac{1}{\sqrt{1 - \frac{8}{9}}} = 3$$

$$\tau_0 = \frac{200}{\ln\left(\frac{10000}{8983}\right) \times 3 \times 10^8 \sqrt{3^2 - 1}} = 2.2 \times 10^{-6} \text{ sec.}$$

Summary

- Lorentz Transformation relates space time coordinates of inertial frames abiding by the postulates of STR

Lorentz's Transformations

$$
\begin{aligned}
x' &= \frac{x - vt}{\sqrt{1 - v^2/c^2}} \\
y' &= y \\
z' &= z \\
t' &= \frac{t - (v/c^2)x}{\sqrt{1 - v^2/c^2}}
\end{aligned}
$$

Inverse Lorentz's Transformations

$$
\begin{aligned}
x &= \frac{x' + vt'}{\sqrt{1 - v^2/c^2}} \\
y &= y' \\
z &= z' \\
t &= \frac{t' + (v/c^2)x'}{\sqrt{1 - v^2/c^2}}
\end{aligned}
$$

- The main consequences of Lorentz transformations are:
  - Length contraction: $l = \frac{l_0}{\gamma}$
  - Time dilation: $T = \gamma\tau$